

CONTACT

khaledmabrouk714@gmail.com
+216 22 959 538
khaledmabrouk.vercel.app
linkedin.com/in/mabrouk-khaled
Tunis, Tunisia
Remote / EU Relocation

CERTIFICATIONS

AWS SAA-C03
Solutions Architect — Associate
AWS Cloud Practitioner (CCP)
Azure Fundamentals (AZ-900)
Azure AI Fundamentals (AI-900)

SKILLS

Back: Node.js / TypeScript, Spring Boot, REST, WebSockets, OCPP, Stripe

Mobile & Front: Flutter (Bloc), React, Angular, Android (Kotlin)

Cloud: GCP (Cloud Run), AWS (Lambda, Cognito), Docker, Kubernetes, Terraform

CI/CD: GitHub Actions, GitLab CI

Architecture: Microservices, Serverless, Event-Driven, Zero-Downtime, 99.9 % uptime

AI / LLM: OpenAI (GPT), Anthropic (Claude), Mistral, NLP, DTW

DB: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

Security: OAuth2 / JWT, RBAC, IAM, GDPR

LANGUAGES

Arabic — *Native*
French — *B2*
English — *B2*

RECOGNITION

Ibda3 Laureate
Univ. Francophonie & ICESCO
DevOps Tunis — active member

KHALED MABROUK

Software Engineer | Cloud-Native Architect

Full-Stack Flutter, React & Node.js • AWS Certified Solutions Architect (SAA-C03)

PROFILE

AWS-certified Software Engineer (SAA-C03) architecting **cloud-native** systems on **GCP/AWS** and shipping full-stack products with **Flutter, React & Node.js** across **four production domains**: **E-Mobility (OCPP)**, **EdTech (LLM)**, **HealthTech (AI)** and **Assistive IoT (NLP/DTW)**. Own the full delivery cycle — architecture → CI/CD → observability — with a sharp focus on **resilience**, **security-by-design** and **measurable business impact**.

PROFESSIONAL EXPERIENCE

Software Engineer & Cloud Architect — Aug 2025 — Present
ZCE — Zero Carbon Emission • Grenoble • Full Remote

- **Built** full-stack the **EV charging supervision & operations applications**: **React/Angular** web, **Flutter (Bloc)** mobile, **Node.js/TypeScript** back-end on **Google Cloud**.
- **Architected** the **GCP microservices (Cloud Run / Cloud SQL)** and core **APIs** (authentication, data) — SLOs, tracing, alerting: **99.9 % availability**.
- **Engineered real-time OCPP** (WebSockets): session start **< 1 s**; **drove feature definition** (requirements, specifications, priorities).
- **Industrialized GitHub Actions CI/CD** (zero-downtime), **cloud spend** **−30 %** (Redis, auto-scaling); established **code review + security-by-design** (RBAC, least-privilege IAM).

Node.js | TypeScript | Flutter (Bloc) | React | Angular | GCP | Redis | Docker | GitHub Actions | OCPP | WebSockets

Full-Stack & AI Engineer — EdTech — Jun 2026 — Present
Pro-Corrector • Freelance • Part-time, alongside current role • Remote

- **Designed and built** a SaaS platform for **AI-automated grading of exam papers** for secondary-school teachers: automatic **segmentation of scanned copies**, **LLM-assisted grading**, per-skill statistics.
- **Architected** the **Node.js/TypeScript + PostgreSQL** back-end and **React/TypeScript** front-end; **multi-LLM orchestration (OpenAI, Anthropic, Mistral)**.
- **Implemented online billing (Stripe)** and **GDPR-by-design** compliance (minors' data).

React | TypeScript | Node.js | PostgreSQL | OpenAI / Anthropic / Mistral | Stripe | GCP

Full-Stack Mobile & Web — HealthTech AI — Oct 2024 — Mar 2025
Maydena • Freelance • Nantes, France • Remote

- **Delivered** an end-to-end **AI medical platform** (Android Kotlin + Angular + Spring Boot) integrating a **clinical prediction engine**.
- **Secured** the backend (**JWT + granular RBAC** patient / practitioner / admin); hybrid **MongoDB + MySQL** persistence optimized for clinical workflows.
- **Shipped** via **GitLab CI/CD** → **Kubernetes** — reproducible environments, friction-free releases.

Android (Kotlin) | Angular | Spring Boot | MongoDB | MySQL | JWT | GitLab CI | Kubernetes

Flutter Mobile & Serverless Backend — Assistive IoT — Mar 2024 — Sep 2024
Gewinner • Capstone • Ariana, Tunisia

- **Built** a **Flutter** voice-controlled wheelchair app — **DTW** robust to noise (error rate −40 %), **real-time geolocation** and an in-app **NLP chatbot**.
- **Architected** a **serverless AWS backend** (Lambda, API Gateway, Cognito, DynamoDB) — pay-per-use scaling, managed auth.

EDUCATION

National Engineering Diploma — Software Engineering — Sep 2022 — Sep 2024
TEK-UP • Tunisia

Applied Bachelor — Electronics & Automation — Sep 2018 — Jun 2021
ISSAT • Sousse, Tunisia